

Cheat sheet - 2nd DSP exam/UFPA

Chapter 4 – Spectral Analysis

Windowing $x_w[n] = x[n]w[n]$ corresponds to *periodic convolution* in frequency-domain:

$$X_w(e^{j\Omega}) = X(e^{j\Omega}) \otimes W(e^{j\Omega}), \quad (1)$$

Hamming window:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, \dots, N-1. \quad (2)$$

“Periodic” *Hann* window

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, \dots, N-1. \quad (3)$$

Important performance metrics of windows: a) width of its main lobe, b) amplitude of its sidelobes, c) rate of sidelobe decay as frequency increases, d) and scalloping loss.

Expressions to investigate bin and non-bin-centered sinusoids, where $\alpha \in \mathbb{R}$ and $\Delta\Omega = 2\pi/N_{\text{fft}}$:

$$x[n] = A \cos\left(\alpha \frac{2\pi}{N_{\text{fft}}} n\right) \quad \text{and} \quad x[n] = A \cos(\alpha \Delta\Omega n). \quad (4)$$

DTFT $X_w(e^{j\Omega})$ of the windowed cosine $x_w[n] = x[n]w[n]$:

$$X_w(e^{j\Omega}) = \frac{A}{2} \left[\frac{\sin\left(\frac{N(\Omega+\Omega_c)}{2}\right)}{\sin\left(\frac{\Omega+\Omega_c}{2}\right)} e^{-j\frac{(\Omega+\Omega_c)(N-1)}{2}} + \frac{\sin\left(\frac{N(\Omega-\Omega_c)}{2}\right)}{\sin\left(\frac{\Omega-\Omega_c}{2}\right)} e^{-j\frac{(\Omega-\Omega_c)(N-1)}{2}} \right]. \quad (5)$$

Leakage is the redistribution of spectral energy caused by windowing a signal.

Picket-fence effect is the phenomenon whereby the DFT/FFT observes the continuous DTFT only at the FFT bin centers.

Scalloping loss is the reduction in the measured amplitude of a spectral component when its frequency lies between two adjacent DFT/FFT bin centers rather than at a bin center. Improving FFT resolution $\Delta f = F_s/N_{\text{fft}}$ by increasing N_{fft} cannot recover the leakage that occurred due to windowing.

Tabela 1: Energy spectral density (ESD) functions. The units of $\mathcal{G}(f)$, $\mathcal{G}(\omega)/(2\pi)$ and $\mathcal{G}(e^{j\Omega})/(2\pi)$ are J/Hz, J/(rad/s) and J/rad, respectively.

Time	ESD definition	Property
Cont.	$\mathcal{G}(f) = X(f) ^2$	$E = \int_{-\infty}^{\infty} \mathcal{G}(f) df$
Cont.	$\mathcal{G}(\omega) = X(\omega) ^2$	$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{G}(\omega) d\omega$
Discr.	$\mathcal{G}(e^{j\Omega}) = X(e^{j\Omega}) ^2$	$E = \frac{1}{2\pi} \int_{<2\pi>} \mathcal{G}(e^{j\Omega}) d\Omega$

PSD of continuous-time periodic deterministic signal:

$$S(f) = \sum_{k=-\infty}^{\infty} |c_k|^2 \delta(f - kF_0), \quad (6)$$

with Fourier Series coefficients c_k and period $T_0 = 1/F_0$. PSD of discrete-time sinusoid $x[n] = A \cos(\Omega_1 n)$:

$$S(e^{j\Omega}) = \sum_{p=-\infty}^{\infty} \frac{\pi A^2}{2} [\delta(\Omega + \Omega_1 + p2\pi) + \delta(\Omega - \Omega_1 + p2\pi)]$$

Tabela 2: Power spectral density (PSD) functions. The units of $S(f)$, $S(\omega)/(2\pi)$ and $S(e^{j\Omega})/(2\pi)$ are W/Hz, W/(rad/s) and W/rad, respectively.

Time	Definition	Main property
Cont.	$S(f) = \mathcal{F}_f\{R(\tau)\}$	$\mathcal{P} = \int_{-\infty}^{\infty} S(f) df$
Cont.	$S(\omega) = \mathcal{F}_\omega\{R(\tau)\}$	$\mathcal{P} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) d\omega$
Discr.	$S(e^{j\Omega}) = \mathcal{F}\{R[\ell]\}$	$\mathcal{P} = \frac{1}{2\pi} \int_{<2\pi>} S(e^{j\Omega}) d\Omega$

For a WSS random process $\mathcal{X}(t)$ with PSD $S_x(f)$, the PSD of $\mathcal{Y}(t) = \mathcal{X}(t)e^{j2\pi f_c t}$ is $S_y(f) = S_x(f - f_c)$.

Mean-square spectrum (MS spectrum):

$$S_{\text{ms}}[k] = |\text{DTFS}\{x[n]\}|^2. \quad (7)$$

If a WSS process $\mathcal{X}(t)$ is the input of a LTI system $\mathcal{X}(t) \rightarrow \boxed{h(t)} \rightarrow \mathcal{Y}(t)$, then $S_y(f) = |H(f)|^2 S_x(f)$. In discrete-time: $S_y(e^{j\Omega}) = |H(e^{j\Omega})|^2 S_x(e^{j\Omega})$.

The *bilateral* PSD of a white noise is

$$S(f) = \mathcal{F}\left\{\frac{\mathcal{N}_0}{2}\delta(\tau)\right\} = \frac{\mathcal{N}_0}{2}, \quad \forall f \quad (8)$$

and its average power over a bandwidth BW is $\mathcal{P} = \mathcal{N}_0 \text{BW}$.

For discrete-time white noise: $S_y(e^{j\Omega}) = \sigma_x^2 |H(e^{j\Omega})|^2$.

Periodogram is an approximation of $S(f)$ given by

$$\hat{S}[k] \stackrel{\text{def}}{=} \frac{1}{N \text{ BW}} |\text{FFT}\{x_N[n]\}|^2. \quad (9)$$

$$\mathcal{P} \approx \Delta f \sum_{k=0}^{N-1} \hat{S}[k], \quad (10)$$

Relation between MS spectrum and PSD:

$$\hat{S}_{\text{ms}}[k] = \mathcal{P}_k = \Delta f \hat{S}[k]. \quad (11)$$

$$\hat{S}(e^{j\Omega}) = \sum_{k=-N+1}^{N-1} \hat{R}[k] e^{-j\Omega n}. \quad (12)$$

Learn to use Welch's method in Matlab:

```
[Sx,w]=pwelch(x>window,Num_overlap,N_fft,Fs,
'twosided')
```

and Python

```
from scipy.signal import welch
w,Sx=welch(x>window,nperseg=len(window),
noverlap=Num_overlap,nfft=N_fft,
window>window,return_onesided=False)
```

Auto-regressive modeling: find prediction-error FIR filter $A(z) = 1 + \sum_{i=1}^P a_i z^{-i} = 1 - \tilde{A}(z)$, where $\tilde{A}(z) = -\sum_{i=1}^P a_i z^{-i}$ provides the predictions

$$\tilde{y}[n] = -\sum_{i=1}^P a_i y[n-i], \quad (13)$$

such that $A(z)$ minimizes the power of the prediction error is $x[n] = y[n] - \tilde{y}[n]$. Having $A(z)$, the estimated PSD is:

$$\hat{S}(e^{j\Omega}) = |H(e^{j\Omega})|^2 = \frac{g^2}{|A(e^{j\Omega})|^2}, \quad (14)$$

Short-time Fourier transform (STFT):

$$X(\tau, f) = \int_{-\infty}^{\infty} x(t)w(t-\tau)e^{-j2\pi ft} dt, \quad (15)$$